

Contact prediction on matched protein complexes

23 frozen FoldBench dimers: 6 development and 17 test targets. Figures show the 17 test targets.
Each model predicts the whole dimer. Both contact classes use the same experimental residue mask.

Test mean R-precision	Within chains	Between chains
MarinFold multichain	44.3%	2.6%
MarinFold default + 10G	45.1%	1.4%
ESMFold2	74.8%	43.9%
AlphaFold3	76.7%	52.9%

Metric: top R pairs, where R is the native contact count in that class. Intra pools both partners.

Ground truth: pyconfind native-only degree ≥ 0.001 ; separation ≥ 6 within a chain; no inter-chain exclusion.

MarinFold ranking: consensus votes from 100 rollouts; T=1, top-p=0.95, full 8192-token context.

Multichain: exp343 step 280154. Default: exp277 step 266344, A + 10G + B, linker omitted from scoring.

ESMFold2: native chains, no MSA, 20 loops / 100 diffusion steps, best of 5 by 0.8 ipTM + 0.2 pTM.

AlphaFold3: native chains, ColabFold paired/unpaired MSAs, no templates, 10 recycles, 1 seed x 5 samples.

AF3 selects its own highest ranking_score. Structure contacts are ranked by pyconfind degree.

Zero scores remain eligible; ties preserve canonical pair order. Random-tie expectations are in the CSV.

Equivalent-sequence chains retain input order for every method; no ground-truth-based chain swapping.

Uncertainty: 10,000 bootstrap samples of 13 homology groups. Same complexes in every comparison.

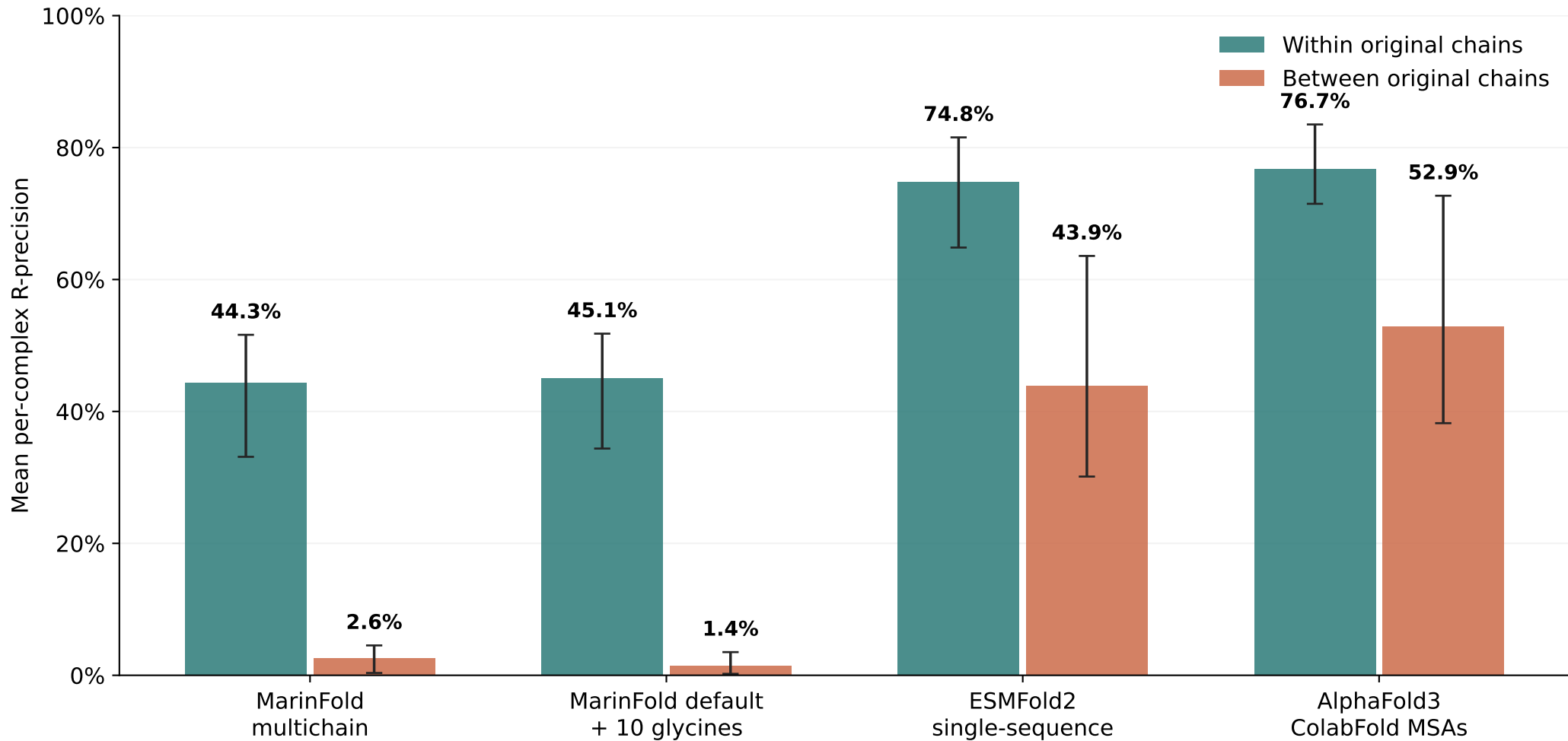
AF3 has MSA information; this is a comparison of these specified pipelines, not a controlled MSA ablation.

Baseline training-set decontamination has not been audited here. Pair holdout was audited for exp343.

AF3: Abramson et al., Nature 630, 493–500 (2024). ESMFold2: Biohub, biohub/ESMFold2.

Generated by: `uv run python plot_four_model_eval.py`

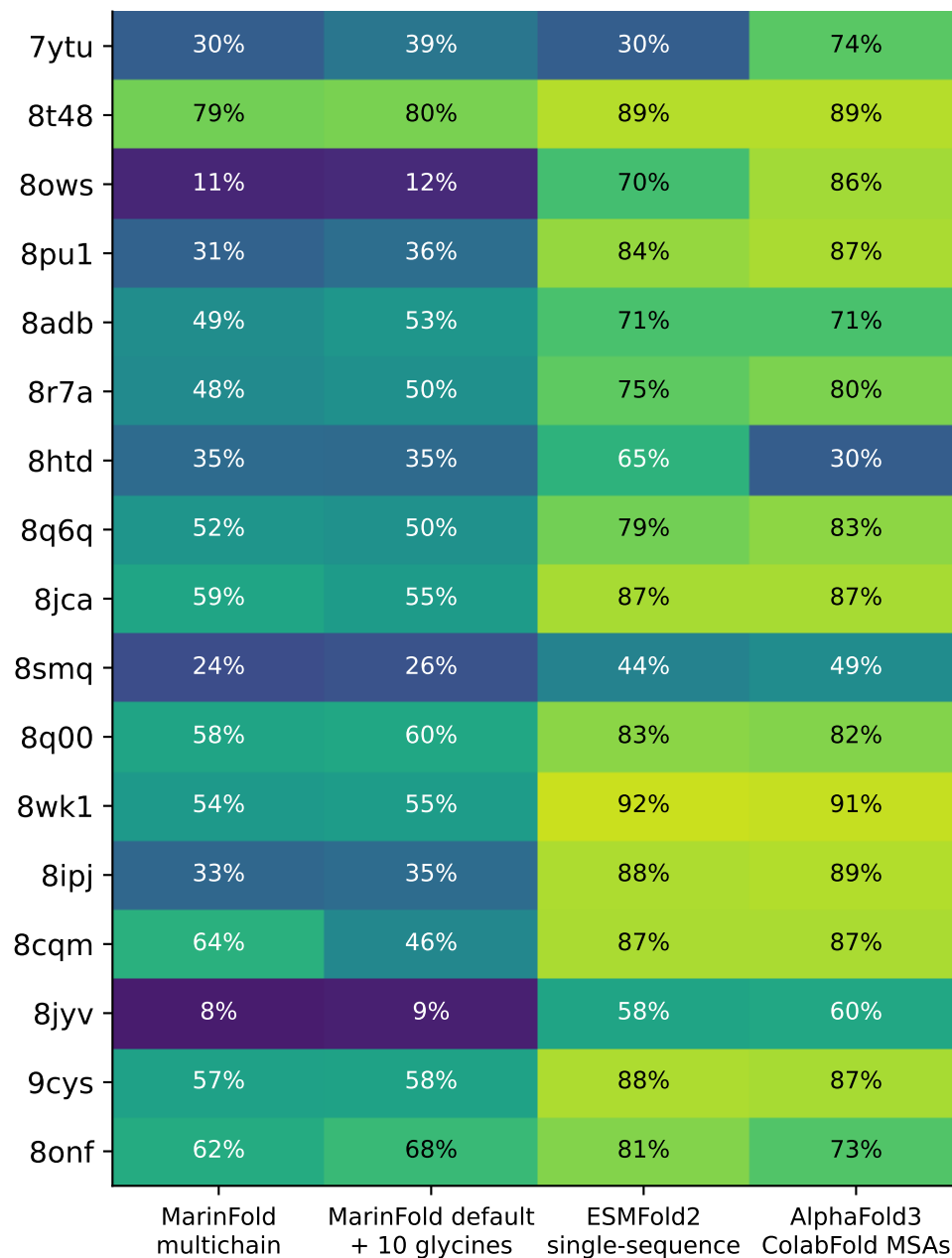
Within-chain and interface accuracy on the same 17 test complexes



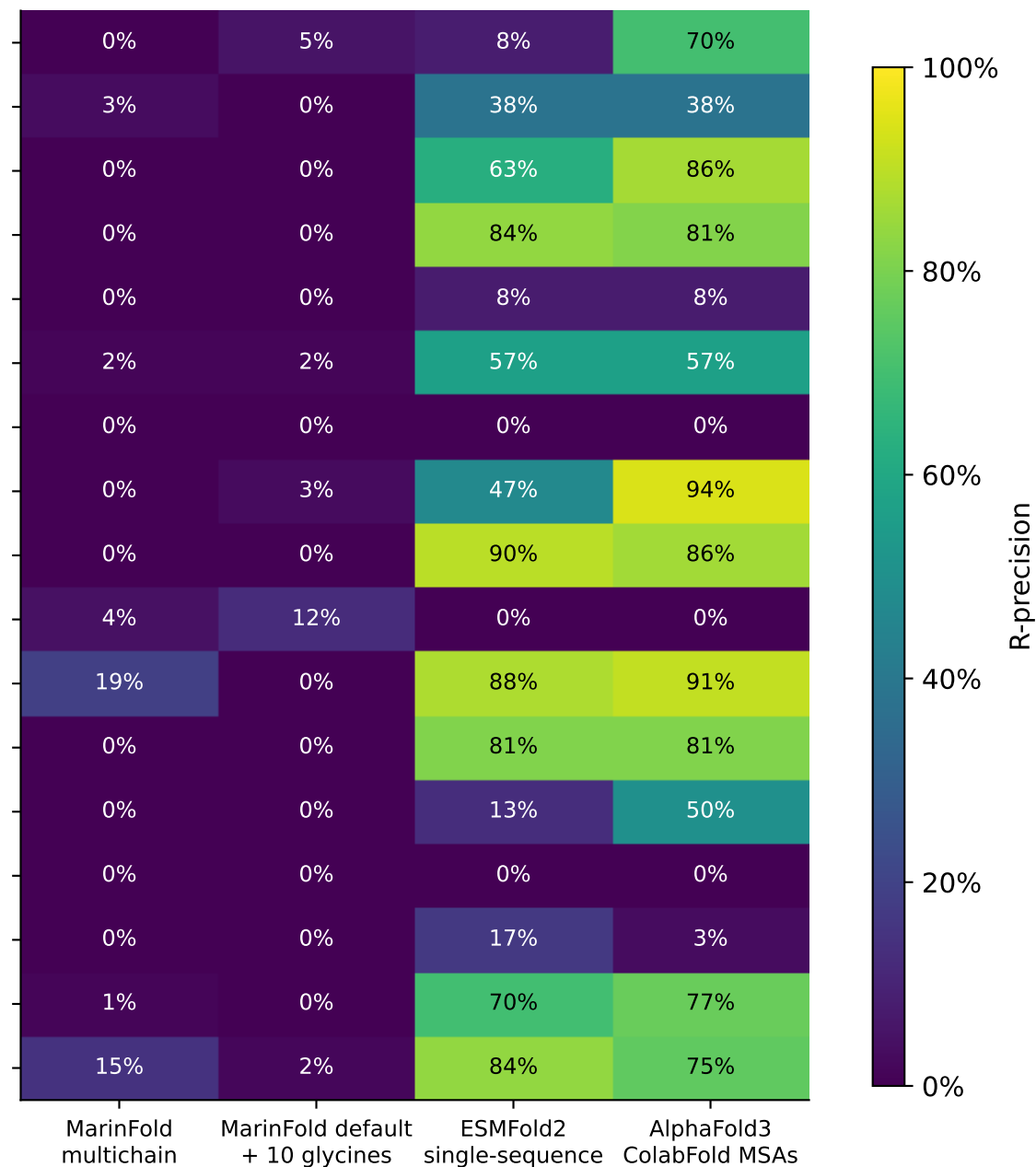
Whiskers: 95% homology-group bootstrap intervals (13 groups). No monomer-only evaluation targets.

Per-complex R-precision · identical test cohort

Within original chains



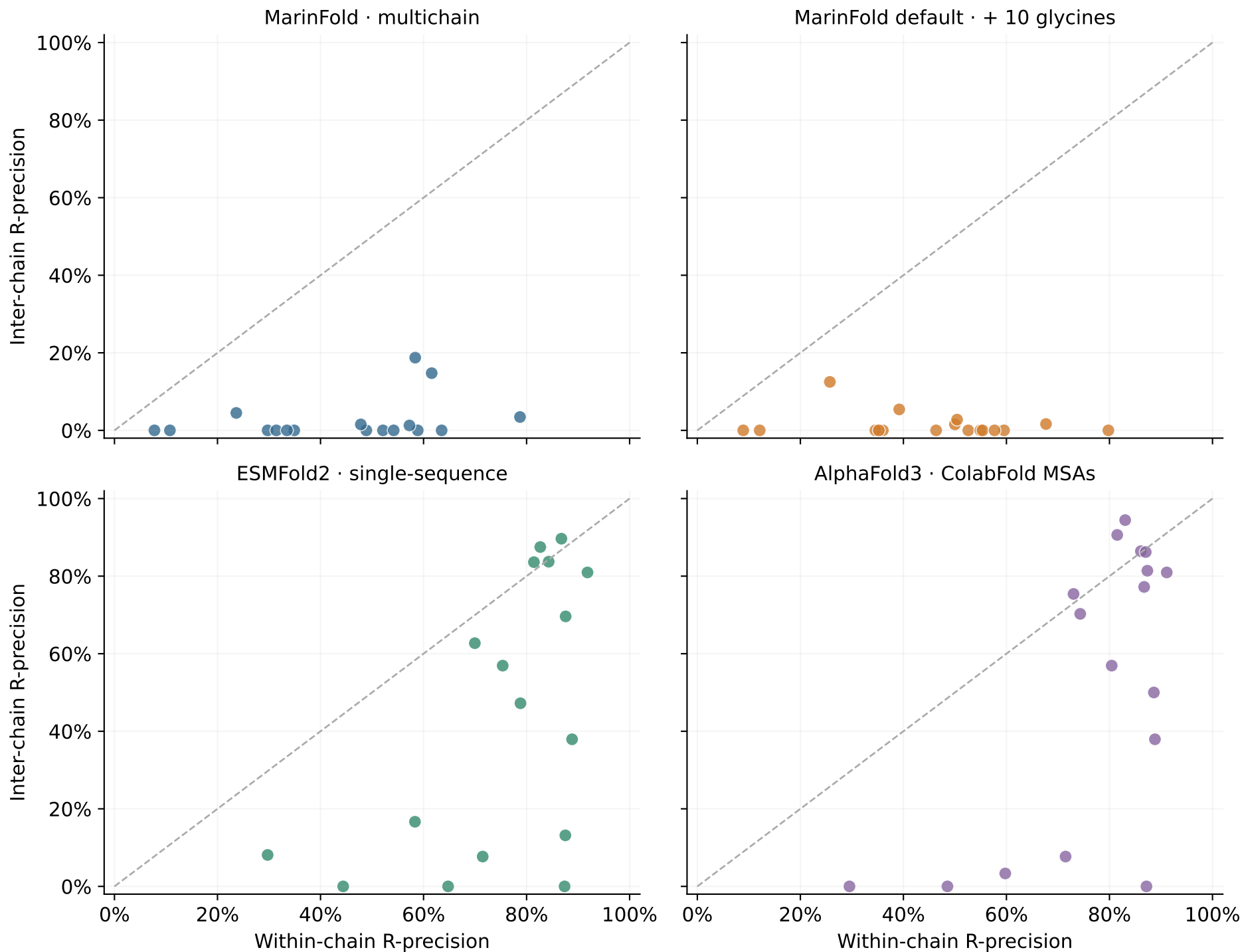
Between original chains



R-precision



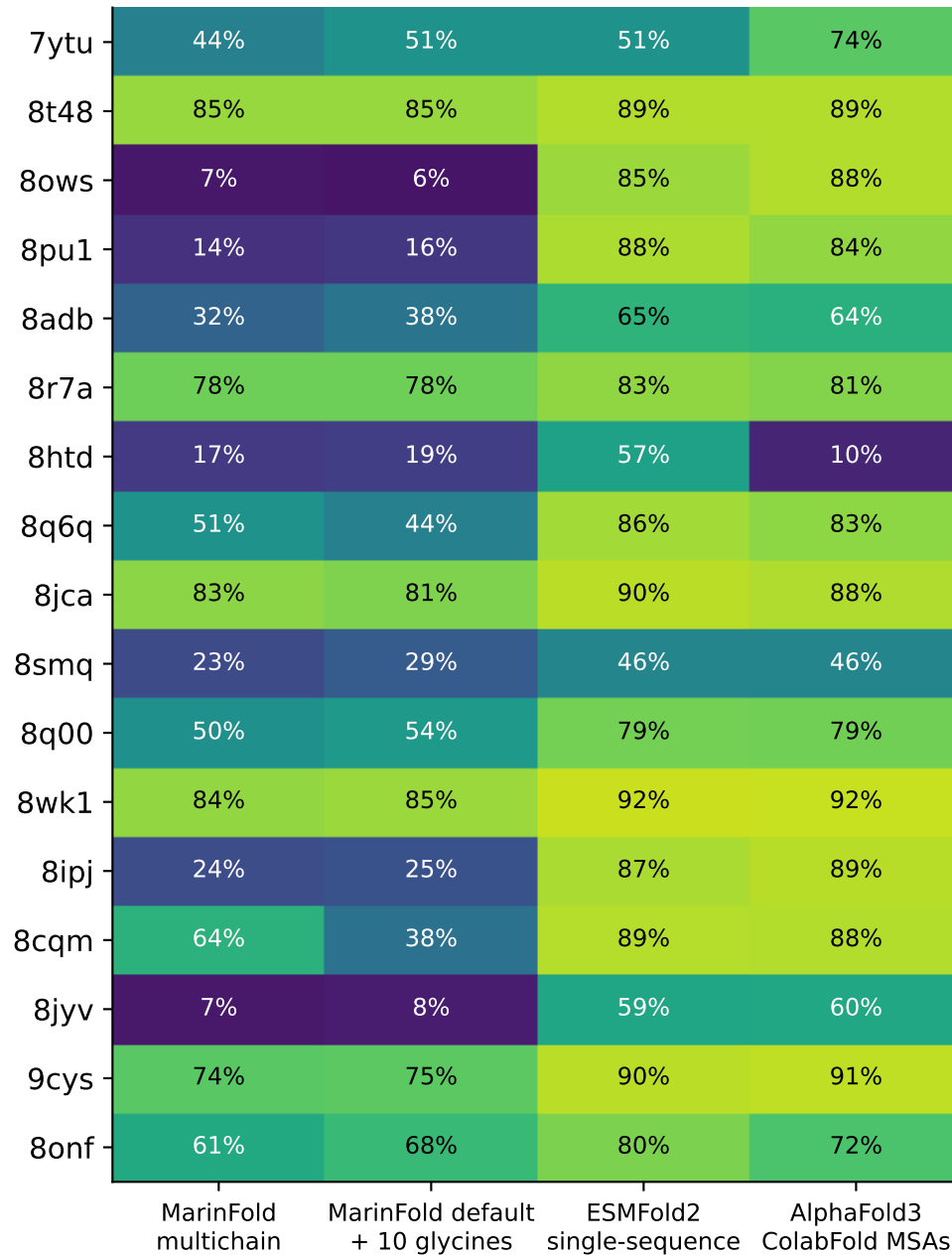
Does a well-predicted pair of chains imply a correct interface?



One point per test complex. The dashed line marks equal intra- and inter-chain accuracy.

Accuracy within each partner, extracted from the same dimer predictions

Original chain A



Original chain B

